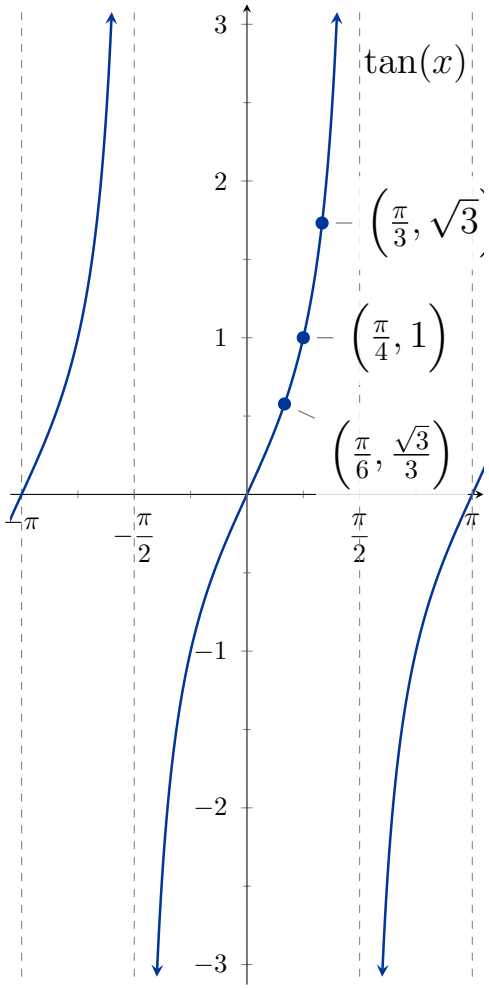
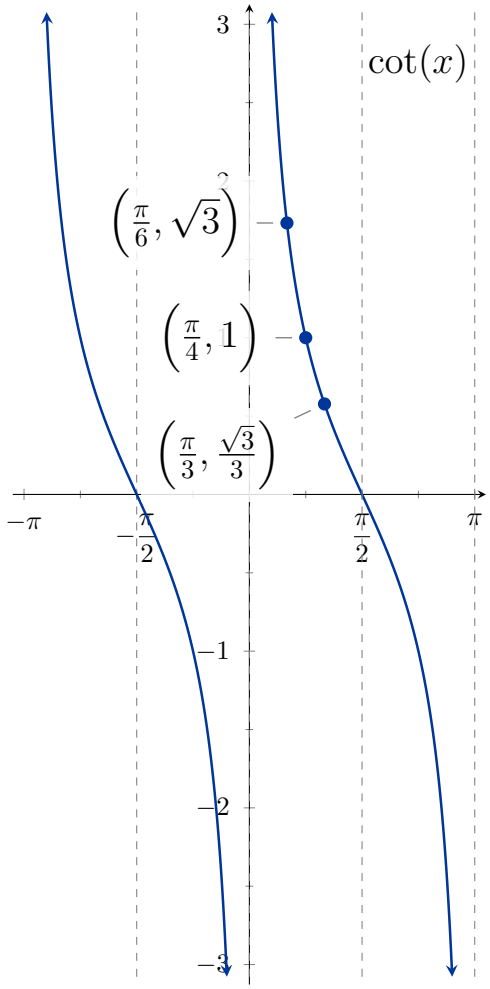
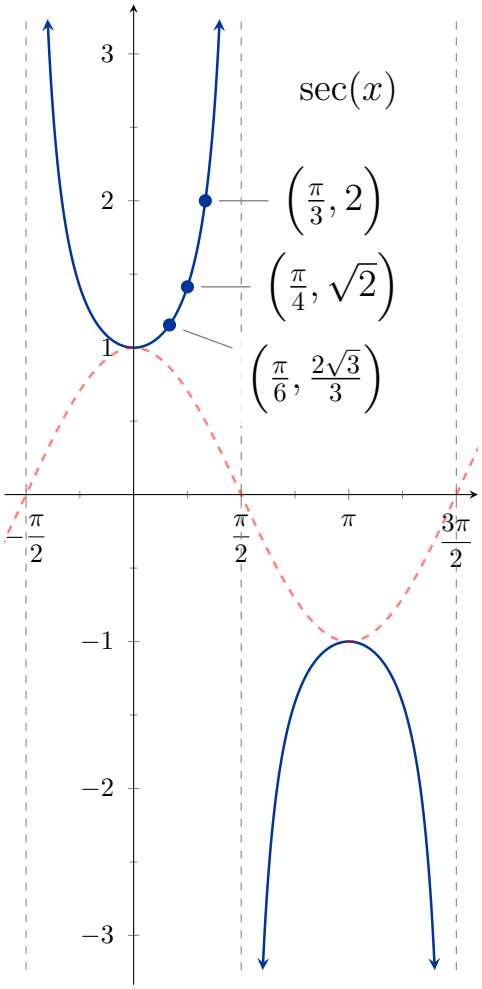
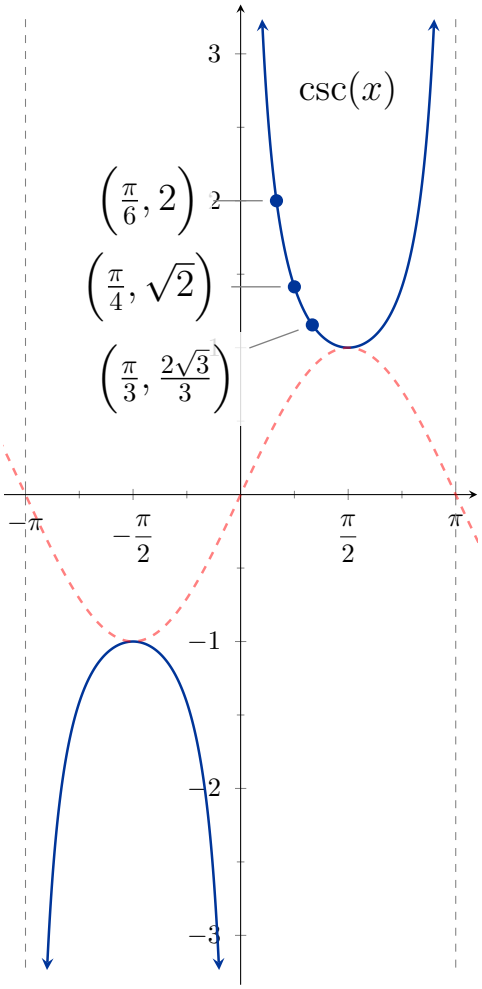
6.2: Graphs of the Other Trigonometric Functions






sec(x)
3
2
1
−1
−2
−3
$\left(\frac{\pi}{3}, 2\right)$
$\left(\frac{\pi}{4}, \sqrt{2}\right)$
$\left(\frac{\pi}{6}, \frac{2\sqrt{3}}{3}\right)$
$-\frac{\pi}{2}$
$\frac{\pi}{2}$
π
$\frac{3\pi}{2}$


csc(x)
3
2
−1
−2
−3
$\left(\frac{\pi}{6}, 2\right)$
$\left(\frac{\pi}{4}, \sqrt{2}\right)$
$\left(\frac{\pi}{3}, \frac{2\sqrt{3}}{3}\right)$
$-\pi$
$-\frac{\pi}{2}$
$\frac{\pi}{2}$
π

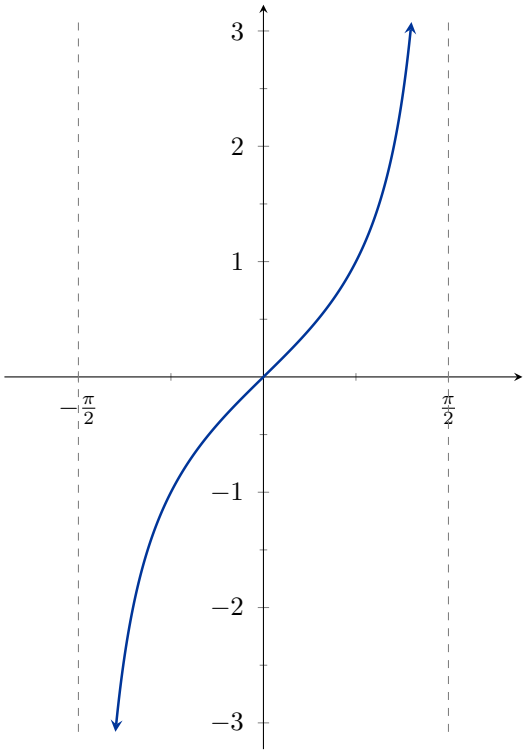
The general forms of the other trigonometric functions are

$$y = A\tan(Bx - C) + D, \qquad y = A\cot(Bx - C) + D,$$
$$y = A\sec(Bx - C) + D, \qquad y = A\csc(Bx - C) + D.$$

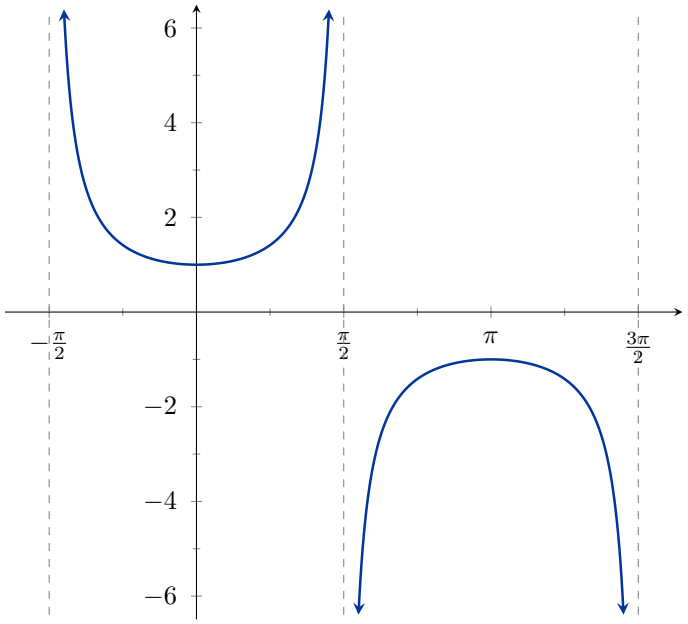
where

- A determines the scaling factor,
- B determines the period,
- C determines the phase shift, and
- D determines the vertical shift.

Example. Draw one period of the function $y = \frac{1}{2}\tan\left(\frac{\pi}{2}x\right)$



Example. Identify the domain, range, and draw one period of $y = 2.5 \sec\left(\frac{2}{5}x\right)$.



Example. Identify the domain, range, and draw one period of $y = -\cot\left(x + \frac{\pi}{2}\right)$.

